

Q1 Company Employee & Department Database

Schema:

- DEPARTMENT (dept_id, dept_name, location)
- EMPLOYEE (emp_id, emp_name, dept_id, designation, salary, manager_id)
- PROJECT (proj_id, proj_name, dept_id)
- WORKS_ON (emp_id, proj_id, hours_worked)

Insert at least 6 departments, 12 employees (some with dept_id as NULL, i.e. unassigned), 5 projects, and 15 WORKS_ON records.

A. Joins (15 Marks)

1. Display employee name, department name, and designation using an INNER JOIN between EMPLOYEE and DEPARTMENT.
2. Using a LEFT JOIN, list all departments along with their employees, including departments that currently have no employees.
3. Using a RIGHT JOIN, list all employees along with their department name, including employees not yet assigned to any department.
4. Write a SELF JOIN on EMPLOYEE to display each employee's name alongside their manager's name.

B. Views (10 Marks)

1. Create a view EMP_DEPT_VIEW showing emp_name, dept_name, designation, salary.
2. Create a view HIGH_EARNERS showing employees earning above ₹50,000, using EMP_DEPT_VIEW as its base (a view built on a view).
3. Create a view DEPT_PROJECT_COUNT showing each department name along with the count of projects it runs.

Q2. University Student, Course & Faculty Database**Schema:**

- STUDENT (student_id, student_name, dept)
- FACULTY (faculty_id, faculty_name, dept)
- COURSE (course_id, course_name, faculty_id, credits)
- ENROLLMENT (enroll_id, student_id, course_id, marks)

Insert at least 8 students, 5 faculty, 6 courses, and 15 enrollment records (include at least one course with no enrollments, and one student not enrolled in any course).

A. Joins (15 Marks)

1. Display student name, course name, and marks using an INNER JOIN across STUDENT, ENROLLMENT, and COURSE.
2. Using a LEFT JOIN, list all students along with the courses they are enrolled in, including students who are enrolled in no course.
3. Using a RIGHT JOIN (or equivalent), list all courses along with enrolled student names, including courses with zero enrollments.
4. Find faculty members who teach courses in which no student has scored above 80 marks (join + aggregate + HAVING).

B. Views (10 Marks)

1. Create a view STUDENT_COURSE_VIEW displaying student_name, course_name, faculty_name, marks.
2. Create a view TOPPERS showing students who scored more than 90 marks in any course, derived from STUDENT_COURSE_VIEW.

3. Create a view FACULTY_WORKLOAD showing faculty name and the number of courses they teach, using GROUP BY inside the view.

4. Query FACULTY_WORKLOAD to find the faculty member with the maximum workload.

Q3. Hospital Patient, Doctor & Appointment Database

Schema:

- DOCTOR (doctor_id, doctor_name, specialization, dept_id)
- PATIENT (patient_id, patient_name, age, gender)
- APPOINTMENT (appt_id, patient_id, doctor_id, appt_date, status)
- DEPARTMENT (dept_id, dept_name)

Insert at least 5 departments, 8 doctors, 10 patients, and 15 appointments (include a patient with no appointments and a doctor with no appointments).

A. Joins (15 Marks)

1. Display patient name, doctor name, and appointment date using an INNER JOIN between PATIENT, APPOINTMENT, and DOCTOR.
2. Using a LEFT JOIN, list all patients with their appointments (if any), including patients who have never booked an appointment.
3. Using a RIGHT JOIN, list all doctors with their appointments, including doctors with no appointments scheduled.
4. Display doctor name, specialization, and department name using a join between DOCTOR and DEPARTMENT.

B. Views (10 Marks)

1. Create a view APPOINTMENT_DETAILS showing patient_name, doctor_name, specialization, appt_date, status.
2. Create a view DOCTOR_APPOINTMENT_COUNT showing doctor name and total number of appointments handled, using GROUP BY.
3. Create a view PENDING_APPOINTMENTS showing only appointments where status = 'Pending', based on APPOINTMENT_DETAILS.
4. Use PENDING_APPOINTMENTS to update (via the base table) the status of a specific appointment to 'Completed', and verify the view reflects the change.

Q4. E-Commerce Customer, Order & Product Database

Schema:

- CUSTOMER (cust_id, cust_name, city)
- PRODUCT (prod_id, prod_name, category, price)
- ORDERS (order_id, cust_id, order_date)
- ORDER_ITEMS (order_id, prod_id, quantity)

Insert at least 6 customers, 10 products, 10 orders, and 20 order-item records (include one customer with no orders and one product never ordered).

A. Joins (15 Marks)

1. Display customer name, order date, product name, and quantity using an INNER JOIN across all four tables.
2. Using a LEFT JOIN, list all customers with their orders, including customers who have never placed an order.
3. Using a RIGHT JOIN (or equivalent), list all products with the orders they appear in, including products never ordered.
4. Find the top-selling product category by total revenue, using multi-table joins with GROUP BY and ORDER BY.

B. Views (10 Marks)

1. Create a view ORDER_SUMMARY showing order_id, cust_name, order_date, total_amount (computed using joins and SUM).
2. Create a view CUSTOMER_SPENDING showing customer name and total amount spent across all orders, using GROUP BY inside the view.
3. Query CUSTOMER_SPENDING to find customers who have spent more than ₹10,000 in total.

1.OUTPUT:

```
mysql> create database companydb;
Query OK, 1 row affected (0.07 sec)

mysql> use companydb;
Database changed
mysql> create table department(dept_id int primary key,dept_name varchar(50),location varchar(50));
Query OK, 0 rows affected (0.16 sec)

mysql> create table employee(emp_id int primary key,emp_name varchar(50),dept_id int,designation varchar(50),salary decimal(10,2),manager_id int,foreign key(dept_id) references department(dept_id));
Query OK, 0 rows affected (0.14 sec)

mysql> create table project(proj_id int primary key,proj_name varchar(50),dept_id int,foreign key(dept_id) references department(dept_id));
Query OK, 0 rows affected (0.10 sec)

mysql> create table works_on(emp_id int,proj_id int,hours_worked int,primary key(emp_id,proj_id),foreign key(emp_id) references employee(emp_id),foreign key(proj_id) references project(proj_id));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into department values(1,'IT','Hyderabad'),(2,'HR','Bangalore'),(3,'Finance','Chennai'),(4,'Marketing','Mumbai'),(5,'Sales','Delhi'),(6,'Research','Pune');
Query OK, 6 rows affected (0.06 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> select * from department;
+-----+-----+-----+
| dept_id | dept_name | location |
+-----+-----+-----+
| 1       | IT        | Hyderabad |
| 2       | HR        | Bangalore |
| 3       | Finance   | Chennai   |
| 4       | Marketing | Mumbai    |
| 5       | Sales     | Delhi     |
| 6       | Research  | Pune      |
+-----+-----+-----+
6 rows in set (0.05 sec)

mysql> select * from department;
+-----+-----+-----+
| dept_id | dept_name | location |
+-----+-----+-----+
| 1       | IT        | Hyderabad |
| 2       | HR        | Bangalore |
| 3       | Finance   | Chennai   |
| 4       | Marketing | Mumbai    |
| 5       | Sales     | Delhi     |
| 6       | Research  | Pune      |
+-----+-----+-----+
6 rows in set (0.05 sec)

mysql> insert into employee values(101,'Rahul',1,'Software Engineer',65000,104),(102,'Priya',1,'Developer',55000,104),(103,'Arjun',2,'HR Executive',45000,106),(104,'Kiran',1,'Project Manager',80000,NULL),(105,'Sneha',4,'Marketing Executive',50000,108),(106,'Anil',2,'HR Manager',70000,NULL),(107,'Divya',5,'Sales Executive',48000,109),(108,'Vikram',4,'Marketing Manager',75000,NULL),(109,'Ravi',5,'Sales Manager',72000,NULL),(110,'Meena',6,'Research Analyst',60000,112),(111,'Asha',NULL,'Intern',25000,NULL),(112,'Varun',NULL,'Research Lead',85000,NULL);
Query OK, 12 rows affected (0.01 sec)
Records: 12 Duplicates: 0 Warnings: 0

mysql> select * from employee;
+-----+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | designation | salary | manager_id |
+-----+-----+-----+-----+-----+-----+
| 101    | Rahul    | 1       | Software Engineer | 65000.00 | 104         |
| 102    | Priya    | 1       | Developer        | 55000.00 | 104         |
| 103    | Arjun    | 2       | HR Executive      | 45000.00 | 106         |
| 104    | Kiran    | 1       | Project Manager   | 80000.00 | NULL        |
| 105    | Sneha    | 4       | Marketing Executive | 50000.00 | 108         |
| 106    | Anil     | 2       | HR Manager        | 70000.00 | NULL        |
| 107    | Divya    | 5       | Sales Executive   | 48000.00 | 109         |
| 108    | Vikram   | 4       | Marketing Manager  | 75000.00 | NULL        |
| 109    | Ravi     | 5       | Sales Manager     | 72000.00 | NULL        |
| 110    | Meena    | 6       | Research Analyst  | 60000.00 | 112         |
| 111    | Asha     | NULL    | Intern            | 25000.00 | NULL        |
| 112    | Varun    | NULL    | Research Lead     | 85000.00 | NULL        |
+-----+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

2.OUTPUT

```
MySQL 8.0 Command Line Client - Unicode

mysql> create database universitydb;
Query OK, 1 row affected (0.14 sec)

mysql> use universitydb;
Database changed
mysql> create table student(student_id int primary key,student_name varchar(50),dept varchar(50));
Query OK, 0 rows affected (0.15 sec)

mysql> insert into student values(1,'Rahul','CSE'),(2,'Priya','CSE'),(3,'Arjun','ECE'),(4,'Sneha','ECE'),(5,'Kiran','IT'),(6,'Divya','IT'),(7,'Ravi','CSE'),(8,'Pranathi','IT');
Query OK, 8 rows affected (0.07 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+-----+-----+
| student_id | student_name | dept |
+-----+-----+-----+
| 1          | Rahul        | CSE  |
| 2          | Priya        | CSE  |
| 3          | Arjun        | ECE  |
| 4          | Sneha        | ECE  |
| 5          | Kiran        | IT   |
| 6          | Divya        | IT   |
| 7          | Ravi         | CSE  |
| 8          | Pranathi     | IT   |
+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> create table faculty(faculty_id int primary key,faculty_name varchar(50),dept varchar(50));
Query OK, 0 rows affected (0.08 sec)

mysql> insert into faculty values(101,'Dr. Kumar','CSE'),(102,'Dr. Priya','ECE'),(103,'Dr. Meena','IT'),(104,'Dr. Ravi','CSE'),(105,'Dr. Anil','IT');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from faculty;
+-----+-----+-----+
| faculty_id | faculty_name | dept |
+-----+-----+-----+
| 101        | Dr. Kumar    | CSE  |
| 102        | Dr. Priya    | ECE  |
| 103        | Dr. Meena    | IT   |
| 104        | Dr. Ravi     | CSE  |
| 105        | Dr. Anil     | IT   |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> create table course(course_id int primary key,course_name varchar(50),faculty_id int,credits int,foreign key(faculty_id) references faculty(faculty_id));
Query OK, 0 rows affected (0.11 sec)

mysql> insert into course values(201,'DBMS',101,4),(202,'Data Structures',101,4),(203,'Computer Networks',104,3),(204,'Digital Electronics',102,4),(205,'Web Technology',103,3),(206,'Database Systems',105,4);
Query OK, 6 rows affected (0.06 sec)
```

```
mysql> insert into course values(201,'DBMS',101,4),(202,'Data Structures',101,4),(203,'Computer Networks',104,3),(204,'Digital Electronics',102,4),(205,'Web Technology',103,3),(206,'Database Systems',105,4);
Query OK, 6 rows affected (0.06 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> select * from course;
+-----+-----+-----+-----+
| course_id | course_name | faculty_id | credits |
+-----+-----+-----+-----+
| 201 | DBMS | 101 | 4 |
| 202 | Data Structures | 101 | 4 |
| 203 | Computer Networks | 104 | 3 |
| 204 | Digital Electronics | 102 | 4 |
| 205 | Web Technology | 103 | 3 |
| 206 | Database Systems | 105 | 4 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> create table enrollment(enroll_id int primary key,student_id int,course_id int,marks int,foreign key(student_id) references student(student_id),foreign key(course_id) references course(course_id));
Query OK, 0 rows affected (0.14 sec)

mysql> insert into enrollment values(1,1,201,95),(2,2,201,88),(3,3,204,76),(4,4,204,82),(5,5,205,91),(6,6,205,79),(7,7,201,85),(8,1,202,92),(9,2,202,81),(10,3,203,75),(11,4,203,68),(12,5,202,94),(13,6,201,73),(14,7,204,89),(15,2,205,87);
Query OK, 15 rows affected (0.05 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> select * from enrollment;
+-----+-----+-----+-----+
| enroll_id | student_id | course_id | marks |
+-----+-----+-----+-----+
| 1 | 1 | 201 | 95 |
| 2 | 2 | 201 | 88 |
| 3 | 3 | 204 | 76 |
| 4 | 4 | 204 | 82 |
| 5 | 5 | 205 | 91 |
| 6 | 6 | 205 | 79 |
| 7 | 7 | 201 | 85 |
| 8 | 1 | 202 | 92 |
| 9 | 2 | 202 | 81 |
| 10 | 3 | 203 | 75 |
| 11 | 4 | 203 | 68 |
| 12 | 5 | 202 | 94 |
| 13 | 6 | 201 | 73 |
| 14 | 7 | 204 | 89 |
| 15 | 2 | 205 | 87 |
+-----+-----+-----+-----+
15 rows in set (0.00 sec)
```

3.OUTPUT

```
MySQL 8.0 Command Line Client - Unicode

mysql> create database hospitaldb;
Query OK, 1 row affected (0.08 sec)

mysql> use hospitaldb;
Database changed
mysql> create table department(dept_id int primary key,dept_name varchar(50));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into department values(1,'Cardiology'),(2,'Neurology'),(3,'Orthopedics'),(4,'Pediatrics'),(5,'Dermatology');
Query OK, 5 rows affected (0.05 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from department;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1 | Cardiology |
| 2 | Neurology |
| 3 | Orthopedics |
| 4 | Pediatrics |
| 5 | Dermatology |
+-----+-----+
5 rows in set (0.00 sec)

mysql> create table doctor(doctor_id int primary key,doctor_name varchar(50),specialization varchar(50),dept_id int,foreign key(dept_id) references department(dept_id));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into doctor values(101,'Dr. Rajesh','Cardiologist',1),(102,'Dr. Priya','Neurologist',2),(103,'Dr. Arun','Orthopedic',3),(104,'Dr. Sneha','Pediatrician',4),(105,'Dr. Kiran','Dermatologist',5),(106,'Dr. Meera','Cardiologist',1),(107,'Dr. Ravi','Neurologist',2),(108,'Dr. Anil','Orthopedic',3);
Query OK, 8 rows affected (0.06 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> select * from doctor;
+-----+-----+-----+-----+
| doctor_id | doctor_name | specialization | dept_id |
+-----+-----+-----+-----+
| 101 | Dr. Rajesh | Cardiologist | 1 |
| 102 | Dr. Priya | Neurologist | 2 |
| 103 | Dr. Arun | Orthopedic | 3 |
| 104 | Dr. Sneha | Pediatrician | 4 |
| 105 | Dr. Kiran | Dermatologist | 5 |
| 106 | Dr. Meera | Cardiologist | 1 |
| 107 | Dr. Ravi | Neurologist | 2 |
| 108 | Dr. Anil | Orthopedic | 3 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> create table patient(patient_id int primary key,patient_name varchar(50),age int,gender varchar(10));
Query OK, 0 rows affected (0.19 sec)

mysql> insert into patient values(1,'Rahul',25,'Male'),(2,'Priya',32,'Female'),(3,'Arjun',45,'Male'),(4,'Sneha',28,'Female'),(5,'Kiran',60,'Male'),(6,'Divya',35,'Female'),(7,'Ravi',50,'Male'),(8,'Meena',42,'Female'),(9,'Asha',22,'Female'),(10,'Varun',55,'Male');
Query OK, 10 rows affected (0.08 sec)
```

```

mysql> insert into patient values(1,'Rahul',25,'Male'),(2,'Priya',32,'Female'),(3,'Arjun',45,'Male'),(4,'Sneha',28,'Female'),(5,'Kiran',60,'Male'),(6,'Divya',
35,'Female'),(7,'Ravi',50,'Male'),(8,'Meena',42,'Female'),(9,'Asha',22,'Female'),(10,'Varun',55,'Male');
Query OK, 10 rows affected (0.08 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> select * from patient;
+-----+-----+-----+-----+
| patient_id | patient_name | age | gender |
+-----+-----+-----+-----+
| 1 | Rahul | 25 | Male |
| 2 | Priya | 32 | Female |
| 3 | Arjun | 45 | Male |
| 4 | Sneha | 28 | Female |
| 5 | Kiran | 60 | Male |
| 6 | Divya | 35 | Female |
| 7 | Ravi | 50 | Male |
| 8 | Meena | 42 | Female |
| 9 | Asha | 22 | Female |
| 10 | Varun | 55 | Male |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> create table appointment(appt_id int primary key,patient_id int,doctor_id int,appt_date date,status varchar(20),foreign key(patient_id) references pati
ent(patient_id),foreign key(doctor_id) references doctor(doctor_id));
Query OK, 0 rows affected (0.11 sec)

mysql> insert into appointment values(1,1,101,'2026-08-01','Completed'),(2,2,102,'2026-08-02','Pending'),(3,3,103,'2026-08-03','Completed'),(4,4,104,'2026-08-
04','Pending'),(5,5,105,'2026-08-05','Completed'),(6,6,101,'2026-08-06','Pending'),(7,7,102,'2026-08-07','Completed'),(8,8,103,'2026-08-08','Pending'),(9,10,1
04,'2026-08-09','Completed'),(10,1,105,'2026-08-10','Pending'),(11,2,101,'2026-08-11','Completed'),(12,3,102,'2026-08-12','Pending'),(13,4,103,'2026-08-13','C
ompleted'),(14,5,104,'2026-08-14','Pending'),(15,6,108,'2026-08-15','Completed');
Query OK, 15 rows affected (0.01 sec)
Records: 15 Duplicates: 0 Warnings: 0

mysql> select * from appointment;
+-----+-----+-----+-----+-----+
| appt_id | patient_id | doctor_id | appt_date | status |
+-----+-----+-----+-----+-----+
| 1 | 1 | 101 | 2026-08-01 | Completed |
| 2 | 2 | 102 | 2026-08-02 | Pending |
| 3 | 3 | 103 | 2026-08-03 | Completed |
| 4 | 4 | 104 | 2026-08-04 | Pending |
| 5 | 5 | 105 | 2026-08-05 | Completed |
| 6 | 6 | 101 | 2026-08-06 | Pending |
| 7 | 7 | 102 | 2026-08-07 | Completed |
| 8 | 8 | 103 | 2026-08-08 | Pending |
| 9 | 10 | 104 | 2026-08-09 | Completed |
| 10 | 1 | 105 | 2026-08-10 | Pending |
| 11 | 2 | 101 | 2026-08-11 | Completed |
| 12 | 3 | 102 | 2026-08-12 | Pending |
| 13 | 4 | 103 | 2026-08-13 | Completed |
| 14 | 5 | 104 | 2026-08-14 | Pending |
| 15 | 6 | 108 | 2026-08-15 | Completed |
+-----+-----+-----+-----+-----+
15 rows in set (0.00 sec)

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4.OUTPUT

```

mysql> create database ecommerce_db;
Query OK, 1 row affected (0.10 sec)

mysql> use ecommerce_db;
Database changed
mysql> create table customer(cust_id int primary key,cust_name varchar(50),city varchar(50));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into customer values(1,'Rahul','Hyderabad'),(2,'Priya','Bangalore'),(3,'Arjun','Chennai'),(4,'Sneha','Mumbai'),(5,'Kiran','Delhi'),(6,'Asha','Pune');
Query OK, 6 rows affected (0.12 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> select * from customer;
+-----+-----+-----+
| cust_id | cust_name | city |
+-----+-----+-----+
| 1       | Rahul     | Hyderabad |
| 2       | Priya     | Bangalore |
| 3       | Arjun     | Chennai   |
| 4       | Sneha     | Mumbai    |
| 5       | Kiran     | Delhi     |
| 6       | Asha      | Pune      |
+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> create table product(prod_id int primary key,prod_name varchar(50),category varchar(50),price decimal(10,2));
Query OK, 0 rows affected (0.13 sec)

mysql> insert into product values(101,'Laptop','Electronics',55000),(102,'Mobile Phone','Electronics',25000),(103,'Headphones','Electronics',3000),(104,'Keyboard','Electronics',2000),(105,'Mouse','Electronics',1200),(106,'Office Chair','Furniture',8000),(107,'Table','Furniture',10000),(108,'Backpack','Accessories',2500),(109,'Smart Watch','Accessories',5000),(110,'Printer','Electronics',15000);
Query OK, 10 rows affected (0.08 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> select * from product;
+-----+-----+-----+-----+
| prod_id | prod_name | category | price |
+-----+-----+-----+-----+
| 101     | Laptop    | Electronics | 55000.00 |
| 102     | Mobile Phone | Electronics | 25000.00 |
| 103     | Headphones | Electronics | 3000.00 |
| 104     | Keyboard  | Electronics | 2000.00 |
| 105     | Mouse     | Electronics | 1200.00 |
| 106     | Office Chair | Furniture | 8000.00 |
| 107     | Table     | Furniture | 10000.00 |
| 108     | Backpack  | Accessories | 2500.00 |
| 109     | Smart Watch | Accessories | 5000.00 |
| 110     | Printer   | Electronics | 15000.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> create table orders(order_id int primary key,cust_id int,order_date date,foreign key(cust_id) references customer(cust_id));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into orders values(1001,1,'2026-08-01'),(1002,2,'2026-08-02'),(1003,3,'2026-08-03'),(1004,4,'2026-08-04'),(1005,5,'2026-08-05'),(1006,1,'2026-08-06'),(1007,2,'2026-08-07'),(1008,3,'2026-08-08'),(1009,4,'2026-08-09'),(1010,5,'2026-08-10');
Query OK, 10 rows affected (0.09 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> select * from orders;
+-----+-----+-----+
| order_id | cust_id | order_date |
+-----+-----+-----+
| 1001     | 1       | 2026-08-01 |
| 1002     | 2       | 2026-08-02 |
| 1003     | 3       | 2026-08-03 |
| 1004     | 4       | 2026-08-04 |
| 1005     | 5       | 2026-08-05 |
| 1006     | 1       | 2026-08-06 |
| 1007     | 2       | 2026-08-07 |
| 1008     | 3       | 2026-08-08 |
| 1009     | 4       | 2026-08-09 |
| 1010     | 5       | 2026-08-10 |
+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> create table order_items(order_id int,prod_id int,quantity int,primary key(order_id,prod_id),foreign key(order_id) references orders(order_id),foreign key(prod_id) references product(prod_id));
Query OK, 0 rows affected (0.13 sec)

mysql> insert into order_items values(1001,101,1),(1001,103,2),(1002,102,1),(1002,105,2),(1003,106,1),(1003,108,2),(1004,101,1),(1004,104,1),(1005,102,2),(1005,110,1),(1006,103,3),(1006,105,2),(1007,101,1),(1007,108,1),(1008,106,2),(1008,107,1),(1009,102,1),(1009,110,2),(1010,104,2),(1010,107,1);
Query OK, 20 rows affected (0.10 sec)
Records: 20 Duplicates: 0 Warnings: 0

mysql> create table order_items2(order_id int,prod_id int,quantity int,primary key(order_id,prod_id),foreign key(order_id) references orders(order_id),foreign key(prod_id) references product(prod_id));
Query OK, 0 rows affected (0.13 sec)

mysql> insert into order_items2 values(1001,101,1),(1001,103,2),(1002,102,1),(1002,105,2),(1003,106,1),(1003,108,2),(1004,101,1),(1004,104,1),(1005,102,2),(1005,110,1),(1006,103,3),(1006,105,2),(1007,101,1),(1007,108,1),(1008,106,2),(1008,107,1),(1009,102,1),(1009,110,2),(1010,104,2),(1010,107,1);
Query OK, 20 rows affected (0.10 sec)
Records: 20 Duplicates: 0 Warnings: 0

mysql> select * from order_items;
+-----+-----+-----+
| order_id | prod_id | quantity |
+-----+-----+-----+
| 1001     | 101     | 1         |
| 1001     | 103     | 2         |
| 1002     | 102     | 1         |
| 1002     | 105     | 2         |
| 1003     | 106     | 1         |
| 1003     | 108     | 2         |
| 1004     | 101     | 1         |
| 1004     | 104     | 1         |
| 1005     | 102     | 2         |
| 1005     | 110     | 1         |
| 1006     | 103     | 3         |
| 1006     | 105     | 2         |
| 1007     | 101     | 1         |
| 1007     | 108     | 1         |
| 1008     | 106     | 2         |
| 1008     | 107     | 1         |
| 1009     | 102     | 1         |
| 1009     | 110     | 2         |
| 1010     | 104     | 2         |
| 1010     | 107     | 1         |
+-----+-----+-----+
20 rows in set (0.00 sec)

```